Variable Listing -- OCP Maintenance AI MVP

> ***Generated from source code*** *-- `tools/models/schemas.py` , `api/database/models.py` , `tools/engines/` , `api/routers/` , and `templates/generate_templates.py` . Covers all enumerations, database columns, API endpoint fields, KPI definitions, the 72 valid Failure Mode combinations, and all 14 Excel template field schemas.*

Section A: Enumeration Types

All `str, Enum` classes defined in `tools/models/schemas.py` , organized by domain.

A.1 Equipment & Hierarchy Domain

A.1.1 `EquipmentCriticality`

OCP criticality grades for equipment ranking.

Value	Description
`AA`	Double-A -- highest criticality
`A+`	A-plus -- very high criticality
`A`	A-grade criticality
`B`	B-grade criticality
`C`	C-grade criticality
`D`	D-grade -- lowest criticality

A.1.2 `EquipmentStatus`

Lifecycle state of a physical asset.

Value	Description
`ACTIVE`	Equipment is in operation
`INACTIVE`	Equipment is idle / mothballed
`DECOMMISSIONED`	Equipment permanently removed

A.1.3 `NodeType`

Level type in the 6-level plant hierarchy.

Value	Level	Description
`PLANT`	1	Top-level facility
`AREA`	2	Functional area within plant
`SYSTEM`	3	Process system
`EQUIPMENT`	4	Physical equipment item
`SUB_ASSEMBLY`	5	Sub-assembly of equipment
`MAINTAINABLE_ITEM`	6	Lowest replaceable component

A.1.4 `EquipmentCategory`

Classification of equipment by type (Module 4 library).

Value	Description
`PUMP`	Centrifugal, positive displacement, etc.
`CONVEYOR`	Belt, screw, chain conveyors
`CRUSHER`	Jaw, cone, gyratory crushers
`MILL`	SAG, ball, rod mills
`FLOTATION_CELL`	Flotation separation
`THICKENER`	Thickening/settling
`DRYER`	Drying equipment
`KILN`	Rotary kilns
`MOTOR`	Electric motors
`TRANSFORMER`	Electrical transformers
`COMPRESSOR`	Gas compressors
`VALVE`	Control / isolation valves
`TANK`	Storage / process tanks
`FILTER`	Filtration equipment
`AGITATOR`	Agitation equipment
`OTHER`	Uncategorised

A.1.5 `ComponentCategory`

Component classification by discipline.

Value	Description
`MECHANICAL`	Mechanical components
`ELECTRICAL`	Electrical components
`INSTRUMENTATION`	Sensors, PLCs, transmitters
`STRUCTURAL`	Frames, housing, supports
`HYDRAULIC`	Hydraulic systems
`PNEUMATIC`	Pneumatic systems

A.1.6 `LibrarySource`

Origin of a library item.

Value	Description
`R8_LIBRARY`	From R8 standard library
`OEM`	From OEM documentation
`CUSTOM`	User-defined
`AI_GENERATED`	Generated by AI engine

A.2 Field Capture Domain

A.2.1 `CaptureType`

Type of field data capture.

Value	Description
`VOICE`	Voice recording transcribed
`TEXT`	Typed text input
`IMAGE`	Photo capture only
`VOICE+IMAGE`	Combined voice + photo

A.2.2 `Language`

Supported languages for field capture.

Value	Description
`fr`	French
`en`	English
`ar`	Arabic

A.2.3 `ResolutionMethod`

How equipment was identified from field input.

Value	Description
`EXACT_MATCH`	Exact TAG/ID match
`FUZZY_MATCH`	Fuzzy text matching
`IMAGE_OCR`	OCR from photo
`MANUAL`	Manually selected

A.2.4 `VisualSeverity`

Severity from image analysis.

Value	Description
`LOW`	Minor visual anomaly
`MEDIUM`	Moderate degradation
`HIGH`	Significant damage
`CRITICAL`	Immediate attention required

A.3 Work Orders Domain

A.3.1 `WorkRequestStatus`

Lifecycle of a structured work request.

Value	Description
`DRAFT`	Initial creation
`PENDING_VALIDATION`	Awaiting planner review
`VALIDATED`	Approved by planner
`REJECTED`	Rejected by planner
`SUBMITTED_TO_SAP`	Sent to SAP PM

A.3.2 `WorkOrderType`

SAP PM order types.

Value	Description
`PM01_INSPECTION`	Inspection order
`PM02_PREVENTIVE`	Preventive maintenance
`PM03_CORRECTIVE`	Corrective maintenance

A.3.3 `Priority`

Work order priority levels.

Value	Description
`1_EMERGENCY`	Immediate response
`2_URGENT`	Response within shift
`3_NORMAL`	Planned within week
`4_PLANNED`	Scheduled per plan

A.3.4 `AvailabilityStatus`

Spare part stock status.

Value	Description
`IN_STOCK`	Available in warehouse
`LOW_STOCK`	Below reorder point
`OUT_OF_STOCK`	Zero on hand
`UNKNOWN`	Status not determined

A.3.5 `BacklogWOType`

Simplified SAP order type for backlog.

Value	Description
`PM01`	Inspection
`PM02`	Preventive
`PM03`	Corrective

A.3.6 `BacklogStatus`

Status of a backlog item.

Value	Description
`AWAITING_MATERIALS`	Blocked on parts
`AWAITING_SHUTDOWN`	Needs shutdown window
`AWAITING_RESOURCES`	No available workforce
`AWAITING_APPROVAL`	Pending authorization
`SCHEDULED`	Placed on schedule
`IN_PROGRESS`	Execution underway

A.3.7 `WOHistoryStatus`

Work order lifecycle states (historical).

Value	Description
`CREATED`	WO created in system
`RELEASED`	WO released for scheduling
`IN_PROGRESS`	Execution started
`COMPLETED`	Work finished
`CLOSED`	Administratively closed
`CANCELLED`	WO cancelled

A.3.8 `MaterialsReadyStatus`

Material readiness for a work package.

Value	Description
`READY`	All parts available
`PARTIAL`	Some parts missing
`NOT_READY`	Parts not available

A.3.9 `AlertType`

Backlog alert categories.

Value	Description
`OVERDUE`	Item past due date
`MATERIAL_DELAY`	Material delivery delayed
`RESOURCE_CONFLICT`	Workforce double-booked
`PRIORITY_ESCALATION`	Priority has been raised

A.3.10 `SAPWorkOrderStatus`

SAP PM work order system statuses.

Value	Description
`PLN`	Planned
`FMA`	Firm
`LPE`	Released for execution
`LIB`	Released
`IMPR`	In process
`NOTP`	Not planned
`NOTI`	Notification
`CTEC`	Technically completed

A.3.11 `SAPNotificationStatus`

SAP PM notification statuses.

Value	Description
`MEAB`	Open
`METR`	In process
`ORAS`	Order assigned
`MECE`	Closed

A.4 Planner & Scheduling Domain

A.4.1 `ShiftType`

Shift definitions.

Value	Description
`MORNING`	Day shift
`AFTERNOON`	Swing shift
`NIGHT`	Night shift

A.4.2 `ShutdownType`

Planned shutdown categories.

Value	Description
`MINOR_8H`	Minor -- up to 8 hours
`MAJOR_20H_PLUS`	Major -- 20+ hours

A.4.3 `RiskLevel`

General risk classification.

Value	Description
`LOW`	Acceptable risk
`MEDIUM`	Monitor and plan
`HIGH`	Action required
`CRITICAL`	Immediate intervention

A.4.4 `PlannerAction`

Actions a planner can take on a recommendation.

Value	Description
`APPROVE`	Accept as-is
`MODIFY`	Change details and approve
`ESCALATE`	Raise to management
`DEFER`	Postpone execution

A.4.5 `PlanStrategy`

Maintenance plan strategy type.

Value	Description
`TIME_BASED`	Calendar/meter interval
`CONDITION_BASED`	Triggered by measurements
`PREDICTIVE`	Data-driven prediction

A.4.6 `FrequencyUnit`

Units for maintenance task frequency.

Value	Description
`HOURS`	Calendar hours
`DAYS`	Calendar days
`WEEKS`	Calendar weeks
`MONTHS`	Calendar months
`YEARS`	Calendar years
`HOURS_RUN`	Running hours
`OPERATING_HOURS`	Operating hours
`TONNES`	Throughput in tonnes
`CYCLES`	Machine cycles

A.4.7 `PlanStatus`

Maintenance plan lifecycle.

Value	Description
`ACTIVE`	Currently in effect
`SUSPENDED`	Temporarily paused
`DELETED`	Permanently removed

A.4.8 `WeeklyProgramStatus`

Weekly scheduling program states.

Value	Description
`DRAFT`	Being assembled
`FINAL`	Locked for execution
`ACTIVE`	Currently executing
`COMPLETED`	Execution finished

A.4.9 `SchedulingDayRole`

Purpose of each day in the weekly scheduling cycle.

Value	Description
`MEETING`	Scheduling coordination meeting
`ADJUSTMENT`	Mid-week adjustments
`FINAL_PROGRAM`	Final program freeze
`EXECUTION`	Execution day

A.4.10 `ShutdownScope`

Scope classification for shutdown windows.

Value	Description
`MINOR_8H`	Minor -- up to 8 hours
`MAJOR_20H_PLUS`	Major -- 20+ hours

A.4.11 `SupportTaskType`

Support activities required alongside work packages.

Value	Description
`LOTO`	Lock-out/Tag-out
`SCAFFOLDING`	Scaffolding erection/removal
`CRANE`	Crane mobilization
`MANLIFT`	Manlift requirement
`GUARD_REMOVAL`	Guard removal/reinstallation
`CLEANING`	Area cleaning
`COMMISSIONING`	Post-work commissioning

A.4.12 `WorkPackageElementType`

Required elements of a complete work package.

Value	Description
`WORK_PERMIT`	Work permit document
`LOTO_CERT`	LOTO certificate
`MATERIAL_WITHDRAWAL`	Material issue slip
`CHECKLIST`	Pre-job checklist
`ATS_ART`	Risk analysis (ATS/ART)
`EXECUTION_PROCEDURE`	Step-by-step procedure
`WORK_ORDER`	SAP work order reference

A.5 Criticality & FMEA Domain

A.5.1 `CriticalityMethod`

Assessment methodology selector.

Value	Description
`FULL_MATRIX`	Full 11-category risk matrix
`SIMPLIFIED`	Simplified assessment

A.5.2 `CriticalityCategory`

Risk consequence categories (R8 method).

Value	Description
`SAFETY`	Personnel safety impact
`HEALTH`	Occupational health impact
`ENVIRONMENT`	Environmental impact
`PRODUCTION`	Production loss
`OPERATING_COST`	Operating cost increase
`CAPITAL_COST`	Capital expenditure
`SCHEDULE`	Schedule impact
`REVENUE`	Revenue loss
`COMMUNICATIONS`	Communication systems
`COMPLIANCE`	Regulatory compliance
`REPUTATION`	Company reputation

A.5.3 `RiskClass`

Resulting risk classification from assessment.

Value	Description
`I_LOW`	Class I -- Low risk
`II_MEDIUM`	Class II -- Medium risk
`III_HIGH`	Class III -- High risk
`IV_CRITICAL`	Class IV -- Critical risk

A.5.4 `ApprovalStatus`

Approval workflow for strategy items.

Value	Description
`DRAFT`	Initial state
`REVIEWED`	Peer-reviewed
`APPROVED`	Management-approved

A.5.5 `FunctionType`

Equipment function classification (RCM).

Value	Description
`PRIMARY`	Main intended function
`SECONDARY`	Supporting function
`PROTECTIVE`	Safety/protection function

A.5.6 `FailureType`

Functional failure degree.

Value	Description
`TOTAL`	Complete loss of function
`PARTIAL`	Degraded performance

A.5.7 `FMStatus`

Failure mode recommendation status.

Value	Description
`RECOMMENDED`	Active and recommended
`REDUNDANT`	Duplicate / not needed

A.5.8 `CriticalityMode`

GFSN vs R8 assessment mode selector.

Value	Description
`R8`	R8 standard method
`GFSN`	GFSN OCP method

A.5.9 `GFSNCriticalityBand`

GFSN criticality classification bands.

Value	Description
`ALTO`	High criticality
`MODERADO`	Moderate criticality
`BAJO`	Low criticality

A.5.10 `GFSNConsequenceCategory`

GFSN consequence assessment dimensions.

Value	Description
`BUSINESS_IMPACT`	Business impact
`OPERATIONAL_COST`	Operational cost impact
`INTERRUPTION`	Service interruption
`SAFETY`	Safety impact
`ENVIRONMENT`	Environmental impact
`RSC`	Regulatory / Social / Community

A.5.11 `PriorityMode`

Priority calculation method selector.

Value	Description
`ADDITIVE`	Simple additive scoring
`GFSN_MATRIX`	GFSN matrix method

A.5.12 `GFSNPriority`

GFSN priority levels.

Value	Description
`ALTO`	High priority
`MODERADO`	Moderate priority
`BAJO`	Low priority

A.6 FMEA / RCM Domain

A.6.1 `Mechanism`

18 valid failure mechanisms from SRC-09.

Value	Description
`ARCS`	Electrical arcing
`BLOCKS`	Blockage / obstruction
`BREAKS_FRACTURE_SEPARATES`	Fracture or mechanical break
`CORRODES`	Corrosion degradation
`CRACKS`	Crack propagation
`DEGRADES`	General degradation
`DISTORTS`	Physical distortion
`DRIFTS`	Parameter drift
`EXPIRES`	End-of-life expiry
`IMMOBILISED`	Seizure / immobilization
`LOOSES_PRELOAD`	Loss of preload / fastener loosening
`OPEN_CIRCUIT`	Electrical open circuit
`OVERHEATS_MELTS`	Overheating or melting
`SEVERS`	Severing / cutting
`SHORT_CIRCUITS`	Electrical short circuit
`THERMALLY_OVERLOADS`	Thermal overload
`WASHES_OFF`	Wash-off / erosion
`WEARS`	Mechanical wear

A.6.2 `Cause`

41 valid failure causes from SRC-09.

Value	Description
`ABRASION`	Abrasive wear
`AGE`	Age-related deterioration
`BREAKDOWN_IN_INSULATION`	Insulation breakdown
`BREAKDOWN_OF_LUBRICATION`	Lubrication failure
`BIO_ORGANISMS`	Biological organisms
`CHEMICAL_ATTACK`	Chemical attack
`CHEMICAL_REACTION`	Chemical reaction
`CONTAMINATION`	Contamination ingress
`CORROSIVE_ENVIRONMENT`	Corrosive environment exposure
`CREEP`	Material creep
`CREVICE`	Crevice corrosion
`CYCLIC_LOADING`	Cyclic fatigue loading
`DISSIMILAR_METALS_CONTACT`	Galvanic corrosion
`ELECTRICAL_ARCING`	Electrical arcing damage
`ELECTRICAL_OVERLOAD`	Electrical overload
`ENTRAINED_AIR`	Entrained air / cavitation
`EXCESSIVE_FLUID_VELOCITY`	Excessive fluid velocity
`EXCESSIVE_PARTICLE_SIZE`	Oversized particles
`EXCESSIVE_TEMPERATURE`	Excessive temperature
`EXPOSURE_TO_ATMOSPHERE`	Atmospheric exposure
`EXPOSURE_TO_HIGH_TEMP_CORROSIVE`	High-temperature corrosive exposure
`EXPOSURE_TO_HIGH_TEMP`	High-temperature exposure
`EXPOSURE_TO_LIQUID_METAL`	Liquid metal contact
`HIGH_TEMP_CORROSIVE`	High-temperature corrosion
`IMPACT_SHOCK_LOADING`	Impact / shock load
`INSUFFICIENT_FLUID_VELOCITY`	Insufficient fluid velocity

Value	Description
`LACK_OF_LUBRICATION`	Lubrication starvation
`LOW_PRESSURE`	Low fluid pressure
`LUBRICANT_CONTAMINATION`	Contaminated lubricant
`MECHANICAL_OVERLOAD`	Mechanical overload
`METAL_TO_METAL_CONTACT`	Dry metal contact
`OFF_CENTER_LOADING`	Misaligned / off-center load
`OVERCURRENT`	Overcurrent condition
`POOR_ELECTRICAL_CONNECTIONS`	Poor electrical connections
`POOR_ELECTRICAL_INSULATION`	Poor electrical insulation
`RADIATION`	Radiation damage
`RELATIVE_MOVEMENT`	Relative movement / fretting
`RUBBING`	Rubbing contact
`STRAY_CURRENT`	Stray electrical current
`THERMAL_OVERLOAD`	Thermal overload
`THERMAL_STRESSES`	Thermal stress cycling
`UNEVEN_LOADING`	Uneven loading
`USE`	Normal use / wear
`VIBRATION`	Vibration-induced loosening

A.6.3 `FailurePattern`

RCM failure-rate curves (Nowlan-Heap).

Value	Description
`A_BATHTUB`	Pattern A -- bathtub curve (infant + wearout)
`B_AGE`	Pattern B -- age-related wearout
`C_FATIGUE`	Pattern C -- gradual fatigue increase
`D_STRESS`	Pattern D -- initial stress then flat
`E_RANDOM`	Pattern E -- constant random failure
`F_EARLY_LIFE`	Pattern F -- infant mortality then flat

A.6.4 `FailureConsequence`

RCM consequence classification.

Value	Description
`HIDDEN_SAFETY`	Hidden failure with safety impact
`HIDDEN_NONSAFETY`	Hidden failure without safety impact
`EVIDENT_SAFETY`	Evident failure with safety impact
`EVIDENT_ENVIRONMENTAL`	Evident failure with environmental impact
`EVIDENT_OPERATIONAL`	Evident failure with production impact
`EVIDENT_NONOPERATIONAL`	Evident failure -- no direct impact

A.6.5 `StrategyType`

RCM maintenance strategy selection.

Value	Description
`CONDITION_BASED`	Condition monitoring / on-condition task
`FIXED_TIME`	Fixed interval replacement / overhaul
`RUN_TO_FAILURE`	No planned maintenance
`FAULT_FINDING`	Periodic hidden-function check
`REDESIGN`	Design change required
`OEM`	Follow OEM recommendation

A.6.6 `FMECAStage`

Progressive FMECA worksheet stages.

Value	Description
`STAGE_1_FUNCTIONS`	Define functions
`STAGE_2_FAILURES`	Identify functional failures
`STAGE_3_EFFECTS`	Analyse failure effects
`STAGE_4_DECISIONS`	RCM decision logic

A.6.7 `FMECAWorksheetStatus`

FMECA worksheet lifecycle.

Value	Description
`DRAFT`	Being developed
`IN_PROGRESS`	Analysis underway
`COMPLETED`	Analysis complete
`APPROVED`	Reviewed and approved

A.6.8 `RPNCategory`

Risk Priority Number classification (FMECA).

Value	Description
`LOW`	RPN in low range
`MEDIUM`	RPN in medium range
`HIGH`	RPN in high range
`CRITICAL`	RPN in critical range

A.6.9 `JustificationCategory`

Justification for strategy changes vs. existing plans.

Value	Description
`MODIFIED`	Existing task modified
`ELIMINATED`	Existing task removed
`FREQUENCY_CHANGE`	Frequency adjusted
`TACTIC_CHANGE`	Strategy/tactic changed
`MAINTAINED`	Kept as-is
`NEW_TASK`	Brand new task added

A.7 Task & Work Package Domain

A.7.1 `TaskType`

Maintenance task classifications.

Value	Description
`INSPECT`	Visual / sensory inspection
`CHECK`	Functional check / measurement
`TEST`	Diagnostic test
`LUBRICATE`	Lubrication task
`CLEAN`	Cleaning task
`REPLACE`	Component replacement
`REPAIR`	Repair activity
`CALIBRATE`	Instrument calibration

A.7.2 `TaskConstraint`

Task execution constraint.

Value	Description
`ONLINE`	Can run while equipment operates
`OFFLINE`	Requires equipment shutdown
`TEST_MODE`	Requires test / bypass mode

A.7.3 `LabourSpecialty`

Workforce trade specialty.

Value	Description
`FITTER`	Mechanical fitter
`ELECTRICIAN`	Electrician
`INSTRUMENTIST`	Instrument technician
`OPERATOR`	Plant operator
`CONMON_SPECIALIST`	Condition monitoring specialist
`LUBRICATOR`	Lubrication technician

A.7.4 `UnitOfMeasure`

Material quantity units.

Value	Description
`EA`	Each (unit)
`L`	Litre
`KG`	Kilogram
`M`	Metre

A.7.5 `BudgetType`

Budget classification for tasks.

Value	Description
`REPAIR`	Repair budget
`REPLACE`	Replacement budget

A.7.6 `WPType`

Work package grouping strategy.

Value	Description
`STANDALONE`	Single independent package
`SUPPRESSIVE`	Grouped to suppress shutdown count
`SEQUENTIAL`	Must execute in sequence

A.7.7 `WPConstraint`

Work package shutdown requirement.

Value	Description
`ONLINE`	No shutdown required
`OFFLINE`	Shutdown required

A.7.8 `WPApprovalStatus`

Work package approval lifecycle.

Value	Description
`DRAFT`	Being assembled
`REVIEWED`	Peer-reviewed
`APPROVED`	Management-approved
`UPLOADED_TO_SAP`	Transferred to SAP PM

A.8 SAP Integration Domain

A.8.1 `SAPUploadStatus`

SAP upload package lifecycle.

Value	Description
`GENERATED`	Package generated by system
`REVIEWED`	Reviewed by planner
`APPROVED`	Approved for upload
`UPLOADED`	Successfully uploaded to SAP

A.9 Spare Parts & Reliability Domain

A.9.1 `MaterialCriticality`

Spare part criticality classification.

Value	Description
`CRITICAL`	Failure causes immediate stoppage
`IMPORTANT`	Failure degrades performance
`STANDARD`	Routine consumable

A.9.2 `SparePartCriticality`

VED analysis classification.

Value	Description
`VITAL`	Production stops without this part
`ESSENTIAL`	Significant impact if unavailable
`DESIRABLE`	Nice-to-have in stock

A.9.3 `ConsumptionClass`

FSN inventory velocity classification.

Value	Description
`FAST_MOVING`	High consumption rate
`SLOW_MOVING`	Low consumption rate
`NON_MOVING`	No movement in review period

A.9.4 `CostClass`

ABC cost-based inventory classification.

Value	Description
`A_HIGH`	High-value items (~20% SKUs, ~80% value)
`B_MEDIUM`	Medium-value items
`C_LOW`	Low-value items (~80% SKUs, ~20% value)

A.9.5 `ShutdownStatus`

Shutdown event lifecycle.

Value	Description
`PLANNED`	Scheduled for future
`IN_PROGRESS`	Currently executing
`COMPLETED`	Successfully finished
`CANCELLED`	Cancelled

A.9.6 `MoCStatus`

Management of Change request lifecycle.

Value	Description
`DRAFT`	Being prepared
`SUBMITTED`	Submitted for review
`REVIEWING`	Under review
`APPROVED`	Approved
`IMPLEMENTING`	Being implemented
`CLOSED`	Implementation verified
`REJECTED`	Rejected

A.9.7 `MoCCategory`

Management of Change classification.

Value	Description
`EQUIPMENT_MODIFICATION`	Physical equipment change
`PROCESS_CHANGE`	Process parameter change
`STRATEGY_CHANGE`	Maintenance strategy change
`MATERIAL_SUBSTITUTION`	Alternative material/part
`PROCEDURE_UPDATE`	Procedure revision

A.9.8 `JackKnifeZone`

Jack-Knife diagram zone classification (REF-13 7.5.4).

Value	Description
`ACUTE`	High frequency, high downtime per event
`CHRONIC`	High frequency, low downtime per event
`COMPLEX`	Low frequency, high downtime per event
`CONTROLLED`	Low frequency, low downtime per event

A.9.9 `DamageMechanism`

RBI damage mechanism classification.

Value	Description
`CORROSION`	General or localised corrosion
`FATIGUE`	Cyclic fatigue
`CREEP`	High-temperature creep
`EROSION`	Erosive wear
`STRESS_CORROSION`	Stress corrosion cracking
`HYDROGEN_DAMAGE`	Hydrogen embrittlement
`OTHER`	Other mechanisms

A.9.10 `InspectionTechnique`

RBI recommended inspection methods.

Value	Description
`VISUAL`	Visual inspection
`ULTRASONIC_THICKNESS`	UT thickness measurement
`MAGNETIC_PARTICLE`	MPI
`DYE_PENETRANT`	DPI / LPI
`RADIOGRAPHY`	Radiographic testing
`EDDY_CURRENT`	Eddy current testing
`ACOUSTIC_EMISSION`	Acoustic emission monitoring

A.10 KPI & Governance Domain

A.10.1 `KPIStatus`

KPI target attainment status.

Value	Description
`ON_TARGET`	Meeting or exceeding target
`BELOW_TARGET`	Below target
`ABOVE_TARGET`	Above target (may be negative for lower-is-better)

A.10.2 `CAPAStatus`

CAPA lifecycle (ISO 55002 10.2).

Value	Description
`OPEN`	Newly created
`IN_PROGRESS`	Actions underway
`CLOSED`	Actions completed
`VERIFIED`	Effectiveness verified

A.10.3 `CAPAType`

Corrective vs Preventive action.

Value	Description
`CORRECTIVE`	Corrective action
`PREVENTIVE`	Preventive action

A.10.4 `PDCAPhase`

Deming PDCA cycle phases.

Value	Description
`PLAN`	Planning phase
`DO`	Execution phase
`CHECK`	Verification phase
`ACT`	Standardization phase

A.10.5 `HealthDimension`

5 dimensions of the Asset Health Index (REF-12 Rec 4).

Value	Description
`CRITICALITY`	Equipment criticality assessment
`BACKLOG_PRESSURE`	Ratio of pending work to capacity
`STRATEGY_COVERAGE`	% failure modes with approved strategy
`CONDITION_STATUS`	Active condition monitoring alerts
`EXECUTION_COMPLIANCE`	On-time execution of planned work

A.10.6 `VarianceLevel`

Multi-plant variance alert severity (REF-12 Rec 7).

Value	Description
`NORMAL`	Within 2 sigma of portfolio mean
`WARNING`	Between 2-3 sigma deviation
`CRITICAL`	Beyond 3 sigma deviation

A.10.7 `ExpertDomain`

Expert specialization domains (Neuro-Arq TMS).

Value	Description
`RELIABILITY`	Reliability engineering
`MECHANICAL`	Mechanical maintenance
`ELECTRICAL`	Electrical maintenance
`INSTRUMENTATION`	Instrumentation & control
`PROCESS`	Process engineering
`SAFETY`	Safety engineering
`PLANNING`	Maintenance planning
`SPARE_PARTS`	Spare parts management

A.10.8 `StakeholderRole`

Stakeholder roles (ISO 55002 4.2).

Value	Description
`MAINTENANCE_MANAGER`	Head of maintenance
`RELIABILITY_ENGINEER`	Reliability specialist
`PLANNER`	Maintenance planner
`TECHNICIAN`	Field technician
`OPERATOR`	Plant operator
`PLANT_MANAGER`	Plant director
`SAFETY_OFFICER`	HSE officer
`PROCUREMENT`	Procurement / stores

A.10.9 `RCALevel`

RCA investigation depth level.

Value	Description
`1`	Level 1 -- Supervisor + operator
`2`	Level 2 -- Cross-functional team
`3`	Level 3 -- Full investigation with external support

A.10.10 `RCAStatus`

RCA analysis lifecycle.

Value	Description
`OPEN`	Newly opened
`UNDER_INVESTIGATION`	Investigation underway
`COMPLETED`	Analysis completed
`REVIEWED`	Final review complete

A.10.11 `EvidenceType`

RCA evidence classification.

Value	Description
`INFERRED`	Logically inferred
`SENSORY`	Directly observed
`HYPOTHESIS`	Hypothesized

A.10.12 `RootCauseLevel`

Root cause depth classification.

Value	Description
`PHYSICAL`	Physical root cause
`HUMAN`	Human error root cause
`LATENT`	Systemic / organizational

A.10.13 `Evidence5PCategory`

5P evidence collection categories.

Value	Description
`PARTS`	Physical parts evidence
`POSITION`	Location / position evidence
`PEOPLE`	Human witness evidence
`PAPERS`	Documentation evidence
`PARADIGMS`	Organizational / cultural patterns

A.10.14 `SolutionQuadrant`

Solution prioritization quadrant.

Value	Description
`HIGH_BENEFIT_LOW_DIFFICULTY`	Quick wins -- implement first
`HIGH_BENEFIT_HIGH_DIFFICULTY`	Strategic projects
`LOW_BENEFIT_LOW_DIFFICULTY`	Nice-to-have
`LOW_BENEFIT_HIGH_DIFFICULTY`	Avoid / deprioritize

A.11 Reporting & Dashboard Domain

A.11.1 `ReportType`

Generated report types.

Value	Description
`WEEKLY_MAINTENANCE`	Weekly maintenance summary
`MONTHLY_KPI`	Monthly KPI report
`QUARTERLY_REVIEW`	Quarterly management review

A.11.2 `NotificationType`

System alert notification types.

Value	Description
`RBI_OVERDUE`	Overdue RBI inspection
`KPI_BREACH`	KPI target breach
`EQUIPMENT_RISK`	Equipment risk escalation
`BACKLOG_AGING`	Aging backlog items
`CAPA_OVERDUE`	Overdue CAPA actions
`MOC_OVERDUE`	Overdue MoC requests

A.11.3 `NotificationLevel`

Alert severity.

Value	Description
`INFO`	Informational
`WARNING`	Warning
`CRITICAL`	Critical

A.11.4 `ExportFormat`

Data export format options.

Value	Description
`EXCEL`	Microsoft Excel (.xlsx)
`CSV`	Comma-separated values
`PDF`	PDF document

A.11.5 `ImportSource`

Import data source types.

Value	Description
`EQUIPMENT_HIERARCHY`	Equipment hierarchy import
`FAILURE_HISTORY`	Failure history import
`MAINTENANCE_PLAN`	Maintenance plan import

A.11.6 `CorrelationType`

Cross-module correlation analysis types.

Value	Description
`CRITICALITY_FAILURES`	Criticality vs failure frequency
`COST_RELIABILITY`	Cost vs reliability metrics
`HEALTH_BACKLOG`	Health score vs backlog pressure

A.11.7 `TrafficLight`

Dashboard traffic light indicators.

Value	Description
`GREEN`	On target
`AMBER`	Warning zone
`RED`	Off target

A.12 Phase 7 Domain

A.12.1 `WorkPackageReadiness`

Assembly readiness status.

Value	Description
`NOT_STARTED`	No elements assembled
`PARTIAL`	Some elements present
`READY`	All elements present
`BLOCKED`	Missing critical element

A.12.2 `ElementReadinessStatus`

Individual element readiness.

Value	Description
`MISSING`	Not yet created
`DRAFT`	Draft version available
`READY`	Final version available
`EXPIRED`	Expired, needs renewal

A.12.3 `SupportTaskStatus`

Support task progress.

Value	Description
`PENDING`	Not yet started
`IN_PROGRESS`	Underway
`COMPLETED`	Done
`SKIPPED`	Not required / skipped

A.12.4 `ConflictResolutionType`

Scheduling conflict resolution strategies.

Value	Description
`RESCHEDULE`	Move to different date/shift
`SPLIT_PACKAGE`	Break package into parts
`ADD_SHIFT`	Add extra shift
`REASSIGN_SPECIALTY`	Use alternate trade
`EXTEND_WINDOW`	Extend shutdown window

Section B: Database Fields

All SQLAlchemy table definitions from `api/database/models.py`.

B.1 `plants`

Field	Type	Constraints	Description
`plant_id`	String(50)	**PK**	SAP Plant code (e.g., `OCP-JFC1`)
`name`	String(200)	NOT NULL	Plant name (English)
`name_fr`	String(200)	default=""	Plant name (French)
`name_ar`	String(200)	default=""	Plant name (Arabic)
`location`	String(200)	default=""	Geographic location

B.2 `hierarchy_nodes`

Field	Type	Constraints	Description
`node_id`	String(50)	**PK**, default=uuid	Unique node identifier
`node_type`	String(30)	NOT NULL	NodeType enum value
`name`	String(200)	NOT NULL	Node name (English)
`name_fr`	String(200)	default=""	Node name (French)
`code`	String(100)	indexed	Technical code
`parent_node_id`	String(50)	**FK** hierarchy_nodes.node_id, nullable	Parent reference (self-referential)
`level`	Integer	NOT NULL	Hierarchy level (1-6)
`plant_id`	String(50)	**FK** plants.plant_id, nullable	Owning plant
`tag`	String(100)	nullable, indexed	Equipment TAG
`criticality`	String(10)	nullable	EquipmentCriticality value
`status`	String(30)	default="ACTIVE"	EquipmentStatus value
`equipment_lib_ref`	String(50)	nullable	Equipment library reference
`component_lib_ref`	String(50)	nullable	Component library reference
`sap_func_loc`	String(100)	nullable	SAP functional location
`sap_equipment_nr`	String(100)	nullable	SAP equipment number
`order`	Integer	default=1	Display order within parent
`metadata_json`	JSON	nullable	NodeMetadata as JSON

Indexes: `ix_hierarchy_nodes_parent` , `ix_hierarchy_nodes_level_type`

B.3 `criticality_assessments`

Field	Type	Constraints	Description
`assessment_id`	String(50)	**PK**, default=uuid	Assessment identifier
`node_id`	String(50)	**FK** hierarchy_nodes.node_id	Assessed node
`assessed_at`	DateTime	default=now	Assessment timestamp
`assessed_by`	String(100)	default="system"	Assessor name
`method`	String(30)	NOT NULL	FULL_MATRIX or SIMPLIFIED
`criteria_scores`	JSON	NOT NULL	list[CriteriaScore]
`probability`	Integer	NOT NULL	1-5 probability rating
`overall_score`	Float	default=0.0	Computed overall risk score
`risk_class`	String(20)	NOT NULL	RiskClass value
`ai_suggested_class`	String(20)	nullable	AI recommendation
`ai_justification`	Text	nullable	AI reasoning
`status`	String(20)	default="DRAFT"	ApprovalStatus value

B.4 `functions`

Field	Type	Constraints	Description
`function_id`	String(50)	**PK**, default=uuid	Function identifier
`node_id`	String(50)	**FK** hierarchy_nodes.node_id	Equipment node
`function_type`	String(20)	NOT NULL	PRIMARY, SECONDARY, PROTECTIVE
`description`	Text	NOT NULL	Function description (English)
`description_fr`	Text	default=""	Function description (French)
`performance_standard`	Text	nullable	Performance criteria
`ai_generated`	Boolean	default=False	AI-generated flag
`status`	String(20)	default="DRAFT"	ApprovalStatus value

B.5 `functional_failures`

Field	Type	Constraints	Description
`failure_id`	String(50)	**PK**, default=uuid	Failure identifier
`function_id`	String(50)	**FK** functions.function_id	Parent function
`failure_type`	String(20)	NOT NULL	TOTAL or PARTIAL
`description`	Text	NOT NULL	Failure description (English)
`description_fr`	Text	default=""	Failure description (French)

B.6 `failure_modes`

Field	Type	Constraints	Description
`failure_mode_id`	String(50)	**PK**, default=uuid	FM identifier
`functional_failure_id`	String(50)	**FK** functional_failures.failure_id	Parent failure
`fm_status`	String(20)	default="RECOMMENDED"	RECOMMENDED or REDUNDANT
`what`	String(200)	NOT NULL	FM description (starts uppercase)
`mechanism`	String(50)	NOT NULL	Mechanism enum value
`cause`	String(50)	NOT NULL	Cause enum value
`failure_pattern`	String(30)	nullable	Nowlan-Heap pattern (A-F)
`failure_consequence`	String(30)	NOT NULL	FailureConsequence value
`is_hidden`	Boolean	default=False	Hidden failure flag
`failure_effect`	JSON	nullable	FailureEffect as JSON
`strategy_type`	String(30)	NOT NULL	StrategyType value
`ai_generated`	Boolean	default=False	AI-generated flag
`ai_confidence`	Float	nullable	AI confidence (0-1)
`existing_task_source`	String(200)	nullable	Source of existing task
`justification_category`	String(30)	nullable	JustificationCategory value

Indexes: `ix_fm_mechanism_cause`

B.7 `maintenance_tasks`

> ***Note:*** *Tasks are standalone, reusable action definitions. They do NOT have frequency or acceptable limits -- those fields belong to the strategy level. Frequency and limits are defined when a task is linked to a failure mode via a maintenance strategy (see Template 14:* `14_maintenance_strategy.xlsx`*). The* `frequency_value` *and* `frequency_unit` *fields in the database are retained for backward compatibility with work packages but are NOT populated during task-only data loading (Template 04).*

Field	Type	Constraints	Description
`task_id`	String(50)	**PK**, default=uuid	Task identifier (reusable across strategies)
`failure_mode_id`	String(50)	**FK** failure_modes.failure_mode_id, nullable	Linked failure mode
`name`	String(72)	NOT NULL	Task name (max 72 chars for SAP)
`name_fr`	String(200)	default=""	Task name (French)
`task_type`	String(20)	NOT NULL	TaskType value
`is_secondary`	Boolean	default=False	Secondary task flag
`acceptable_limits`	Text	nullable	Acceptance criteria (strategy-level field)
`conditional_comments`	Text	nullable	Conditional notes (strategy-level field)
`consequences`	Text	default=""	Failure consequences
`justification`	Text	nullable	Task justification
`constraint`	String(20)	NOT NULL	ONLINE, OFFLINE, TEST_MODE
`access_time_hours`	Float	default=0.0	Access/setup time
`frequency_value`	Float	NOT NULL	Interval value (set at strategy/WP level)
`frequency_unit`	String(30)	NOT NULL	FrequencyUnit value (set at strategy/WP level)
`origin`	String(200)	nullable	Task origin reference
`budget_type`	String(20)	nullable	NOT_BUDGETED, REPAIR, or REPLACE
`budgeted_life`	Float	nullable	Budgeted life value
`labour_resources`	JSON	nullable	list[LabourResource]
`material_resources`	JSON	nullable	list[MaterialResource]
`tools_list`	JSON	nullable	Required tools

Field	Type	Constraints	Description
`special_equipment`	JSON	nullable	Special equipment needed
`ai_generated`	Boolean	default=False	AI-generated flag
`ai_confidence`	Float	nullable	AI confidence (0-1)
`status`	String(20)	default="DRAFT"	ApprovalStatus value

B.8 `work_packages`

Field	Type	Constraints	Description
`work_package_id`	String(50)	**PK**, default=uuid	WP identifier
`name`	String(40)	NOT NULL	WP name (ALL CAPS, max 40)
`code`	String(50)	indexed	WP code
`node_id`	String(50)	**FK** hierarchy_nodes.node_id	Equipment node
`frequency_value`	Float	NOT NULL	Interval value
`frequency_unit`	String(30)	NOT NULL	FrequencyUnit value
`constraint`	String(20)	NOT NULL	ONLINE or OFFLINE
`access_time_hours`	Float	default=0.0	Access/setup time
`work_package_type`	String(20)	NOT NULL	STANDALONE, SUPPRESSIVE, SEQUENTIAL
`job_preparation`	Text	nullable	Pre-job preparation notes
`post_shutdown`	Text	nullable	Post-shutdown activities
`allocated_tasks`	JSON	nullable	list[AllocatedTask]
`labour_summary`	JSON	nullable	LabourSummary
`material_summary`	JSON	nullable	list[MaterialSummaryEntry]
`sap_upload_ref`	JSON	nullable	SAPUploadRef
`status`	String(20)	default="DRAFT"	WPApprovalStatus value

B.9 `sap_upload_packages`

Field	Type	Constraints	Description
`package_id`	String(50)	**PK**, default=uuid	Package identifier
`generated_at`	DateTime	default=now	Generation timestamp
`plant_code`	String(50)	NOT NULL	SAP plant code
`maintenance_plan`	JSON	nullable	SAPMaintenancePlan
`maintenance_items`	JSON	nullable	list[SAPMaintenanceItem]
`task_lists`	JSON	nullable	list[SAPTaskList]
`status`	String(20)	default="GENERATED"	SAPUploadStatus value

B.10 `work_orders`

Field	Type	Constraints	Description
`work_order_id`	String(50)	**PK**	WO identifier
`order_type`	String(10)	NOT NULL	PM01, PM02, PM03
`equipment_id`	String(100)	indexed	Equipment reference
`equipment_tag`	String(100)	default=""	Equipment TAG
`priority`	String(10)	NOT NULL	Priority (1-4)
`status`	String(20)	NOT NULL	WOHistoryStatus value
`created_date`	Date	NOT NULL	Creation date
`actual_duration_hours`	Float	nullable	Actual work duration
`description`	Text	default=""	WO description
`materials_consumed`	JSON	nullable	list[MaterialConsumed]

B.11 `health_scores`

Field	Type	Constraints	Description
`score_id`	String(50)	**PK**, default=uuid	Score identifier
`node_id`	String(50)	**FK** hierarchy_nodes.node_id	Assessed node
`plant_id`	String(50)	NOT NULL	Plant reference
`equipment_tag`	String(100)	NOT NULL	Equipment TAG
`calculated_at`	DateTime	default=now	Calculation timestamp
`dimensions`	JSON	nullable	list[HealthScoreDimension]
`composite_score`	Float	default=0.0	Weighted composite (0-100)
`health_class`	String(20)	default="UNKNOWN"	HEALTHY, AT_RISK, CRITICAL, UNKNOWN
`trend`	String(20)	default=""	IMPROVING, STABLE, DEGRADING
`recommendations`	JSON	nullable	Action recommendations

B.12 `kpi_metrics`

Field	Type	Constraints	Description
`metrics_id`	String(50)	**PK**, default=uuid	Metrics identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`equipment_id`	String(100)	nullable	Equipment reference (null = plant-level)
`period_start`	Date	NOT NULL	Period start
`period_end`	Date	NOT NULL	Period end
`calculated_at`	DateTime	default=now	Calculation timestamp
`mtbf_days`	Float	nullable	Mean Time Between Failures (days)
`mttr_hours`	Float	nullable	Mean Time To Repair (hours)
`availability_pct`	Float	nullable	Availability (0-100%)
`oee_pct`	Float	nullable	Overall Equipment Effectiveness (0-100%)
`schedule_compliance_pct`	Float	nullable	Schedule compliance (0-100%)
`backlog_hours`	Float	default=0.0	Pending backlog hours
`pm_compliance_pct`	Float	nullable	PM compliance (0-100%)
`total_work_orders`	Integer	default=0	Total WO count
`corrective_wo_count`	Integer	default=0	Corrective WO count
`preventive_wo_count`	Integer	default=0	Preventive WO count
`reactive_ratio_pct`	Float	nullable	Corrective / Total ratio (0-100%)

B.13 `failure_predictions`

Field	Type	Constraints	Description
`prediction_id`	String(50)	**PK**, default=uuid	Prediction identifier
`equipment_id`	String(100)	indexed	Equipment reference
`equipment_tag`	String(100)	NOT NULL	Equipment TAG
`predicted_at`	DateTime	default=now	Prediction timestamp
`weibull_params`	JSON	nullable	WeibullParameters (beta, eta, gamma, r_squared)
`current_age_days`	Float	NOT NULL	Current age in days
`reliability_current`	Float	NOT NULL	R(t) at current age (0-1)
`predicted_failure_window_days`	Float	NOT NULL	Days until predicted failure
`confidence_level`	Float	default=0.9	Confidence level (0.5-0.99)
`risk_score`	Float	default=0.0	Risk score (0-100)
`failure_pattern`	String(30)	nullable	FailurePattern value
`recommendation`	Text	default=""	Action recommendation
`status`	String(20)	default="DRAFT"	ApprovalStatus value

B.14 `capa_items`

Field	Type	Constraints	Description
`capa_id`	String(50)	**PK**, default=uuid	CAPA identifier
`capa_type`	String(20)	NOT NULL	CORRECTIVE or PREVENTIVE
`title`	String(300)	NOT NULL	CAPA title
`description`	Text	NOT NULL	CAPA description
`plant_id`	String(50)	NOT NULL	Plant reference
`equipment_id`	String(100)	nullable	Equipment reference
`source`	String(100)	NOT NULL	Origin (audit, incident, etc.)
`root_cause`	Text	nullable	Root cause description
`current_phase`	String(10)	default="PLAN"	PDCAPhase value
`status`	String(20)	default="OPEN"	CAPAStatus value
`assigned_to`	String(100)	default=""	Responsible person
`created_at`	DateTime	default=now	Creation timestamp
`target_date`	Date	nullable	Due date
`closed_at`	DateTime	nullable	Closure timestamp
`verified_at`	DateTime	nullable	Verification timestamp
`actions_planned`	JSON	nullable	Planned action items
`actions_completed`	JSON	nullable	Completed action items
`effectiveness_verified`	Boolean	default=False	Verification flag
`lessons_learned`	Text	nullable	Lessons learned text

B.15 `variance_alerts`

Field	Type	Constraints	Description
`alert_id`	String(50)	**PK**, default=uuid	Alert identifier
`plant_id`	String(50)	NOT NULL	Alerting plant
`plant_name`	String(200)	NOT NULL	Plant name
`metric_name`	String(100)	NOT NULL	KPI metric name
`plant_value`	Float	NOT NULL	Plant metric value
`portfolio_mean`	Float	NOT NULL	Portfolio average
`portfolio_std`	Float	NOT NULL	Portfolio standard deviation
`z_score`	Float	NOT NULL	Z-score deviation
`variance_level`	String(20)	NOT NULL	NORMAL, WARNING, CRITICAL
`detected_at`	DateTime	default=now	Detection timestamp
`message`	Text	default=""	Alert message

B.16 `expert_cards`

Field	Type	Constraints	Description
`expert_id`	String(50)	**PK**, default=uuid	Expert card identifier
`user_id`	String(100)	indexed	User account ID
`name`	String(200)	NOT NULL	Expert name
`role`	String(50)	NOT NULL	StakeholderRole value
`plant_id`	String(50)	NOT NULL	Home plant
`domains`	JSON	nullable	list[ExpertDomain]
`equipment_expertise`	JSON	nullable	Equipment TAGs known
`certifications`	JSON	nullable	Certification list
`years_experience`	Integer	default=0	Years of experience
`resolution_count`	Integer	default=0	Issues resolved count
`last_active`	DateTime	nullable	Last activity timestamp
`contact_method`	String(200)	default=""	Preferred contact
`languages`	JSON	nullable	Spoken languages

B.17 `audit_log`

Field	Type	Constraints	Description
`id`	Integer	**PK**, autoincrement	Log entry ID
`entity_type`	String(50)	indexed	Entity type name
`entity_id`	String(50)	nullable	Entity identifier
`action`	String(20)	NOT NULL	CREATE, UPDATE, DELETE, APPROVE
`payload`	JSON	nullable	Change payload
`user`	String(100)	default="system"	Acting user
`timestamp`	DateTime	default=now, indexed	Event timestamp

B.18 `field_captures`

Field	Type	Constraints	Description
`capture_id`	String(50)	**PK**, default=uuid	Capture identifier
`technician_id`	String(100)	NOT NULL	Technician user ID
`capture_type`	String(20)	NOT NULL	CaptureType value
`language`	String(5)	default="en"	Detected language
`raw_text`	Text	nullable	Text input content
`raw_voice_text`	Text	nullable	Voice transcription
`images`	JSON	nullable	list[CaptureImage]
`equipment_tag_manual`	String(100)	nullable	Manually-entered TAG
`location_hint`	String(200)	nullable	Location description
`created_at`	DateTime	default=now	Capture timestamp

B.19 `work_requests`

Field	Type	Constraints	Description
`request_id`	String(50)	**PK**, default=uuid	Request identifier
`source_capture_id`	String(50)	**FK** field_captures.capture_id, nullable	Source capture
`status`	String(30)	default="DRAFT"	WorkRequestStatus value
`equipment_id`	String(100)	NOT NULL	Equipment reference
`equipment_tag`	String(100)	NOT NULL	Equipment TAG
`equipment_confidence`	Float	default=0.0	ID confidence (0-1)
`resolution_method`	String(30)	default="MANUAL"	ResolutionMethod value
`problem_description`	JSON	nullable	ProblemDescription
`ai_classification`	JSON	nullable	AIClassification
`spare_parts`	JSON	nullable	list[SuggestedSparePart]
`image_analysis`	JSON	nullable	ImageAnalysis
`validation`	JSON	nullable	Validation
`created_at`	DateTime	default=now	Creation timestamp

Indexes: `ix_work_requests_status`

B.20 `planner_recommendations`

Field	Type	Constraints	Description
`recommendation_id`	String(50)	**PK**, default=uuid	Recommendation identifier
`work_request_id`	String(50)	**FK** work_requests.request_id	Linked work request
`resource_analysis`	JSON	nullable	ResourceAnalysis
`scheduling_suggestion`	JSON	nullable	SchedulingSuggestion
`risk_assessment`	JSON	nullable	RiskAssessment
`planner_action`	String(20)	NOT NULL	PlannerAction value
`ai_confidence`	Float	default=0.0	AI confidence (0-1)
`generated_at`	DateTime	default=now	Generation timestamp

B.21 `backlog_items`

Field	Type	Constraints	Description
`backlog_id`	String(50)	**PK**, default=uuid	Backlog item identifier
`work_request_id`	String(50)	**FK** work_requests.request_id, nullable	Source work request
`equipment_id`	String(100)	NOT NULL	Equipment reference
`equipment_tag`	String(100)	NOT NULL	Equipment TAG
`priority`	String(20)	NOT NULL	Priority value
`wo_type`	String(10)	NOT NULL	PM01, PM02, PM03
`status`	String(30)	default="AWAITING_APPROVAL"	BacklogStatus value
`blocking_reason`	String(200)	nullable	Reason for block
`estimated_hours`	Float	default=4.0	Estimated duration
`specialties`	JSON	nullable	Required specialties
`materials_ready`	Boolean	default=False	Materials available flag
`shutdown_required`	Boolean	default=False	Shutdown needed flag
`age_days`	Integer	default=0	Age in backlog
`created_at`	DateTime	default=now	Creation timestamp

Indexes: `ix_backlog_items_priority` , `ix_backlog_items_status`

B.22 `optimized_backlogs`

Field	Type	Constraints	Description
`optimization_id`	String(50)	**PK**, default=uuid	Optimization identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`period_start`	Date	NOT NULL	Period start
`period_end`	Date	NOT NULL	Period end
`total_items`	Integer	default=0	Total backlog items
`stratification`	JSON	nullable	BacklogStratification
`work_packages`	JSON	nullable	list[BacklogWorkPackage]
`schedule`	JSON	nullable	list[ScheduleEntry]
`alerts`	JSON	nullable	list[BacklogAlert]
`status`	String(20)	default="DRAFT"	Status
`generated_at`	DateTime	default=now	Generation timestamp

B.23 `workforce`

Field	Type	Constraints	Description
`worker_id`	String(50)	**PK**, default=uuid	Worker identifier
`name`	String(200)	NOT NULL	Worker name
`specialty`	String(50)	NOT NULL	LabourSpecialty value
`shift`	String(20)	NOT NULL	ShiftType value
`plant_id`	String(50)	NOT NULL	Plant reference
`available`	Boolean	default=True	Availability flag
`certifications`	JSON	nullable	Certifications list

B.24 `shutdown_calendar`

Field	Type	Constraints	Description
`shutdown_id`	String(50)	**PK**, default=uuid	Shutdown identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`start_date`	Date	NOT NULL	Start date
`end_date`	Date	NOT NULL	End date
`shutdown_type`	String(30)	NOT NULL	MINOR_8H or MAJOR_20H_PLUS
`areas`	JSON	nullable	Affected areas
`description`	Text	default=""	Description

B.25 `inventory_items`

Field	Type	Constraints	Description
`item_id`	String(50)	**PK**, default=uuid	Item identifier
`material_code`	String(50)	indexed	SAP material code
`warehouse_id`	String(50)	default="WH-01"	Warehouse ID
`description`	String(300)	NOT NULL	Part description
`quantity_on_hand`	Integer	default=0	On-hand quantity
`quantity_reserved`	Integer	default=0	Reserved quantity
`quantity_available`	Integer	default=0	Available quantity
`min_stock`	Integer	default=0	Minimum stock level
`reorder_point`	Integer	default=0	Reorder trigger point
`last_movement_date`	Date	nullable	Last stock movement

B.26 `weekly_programs`

Field	Type	Constraints	Description
`program_id`	String(50)	**PK**, default=uuid	Program identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`week_number`	Integer	NOT NULL	ISO week number
`year`	Integer	NOT NULL	Year
`status`	String(20)	default="DRAFT"	WeeklyProgramStatus value
`work_packages`	JSON	nullable	Scheduled work packages
`total_hours`	Float	default=0.0	Total planned hours
`resource_slots`	JSON	nullable	list[ResourceSlot]
`conflicts`	JSON	nullable	list[ResourceConflict]
`support_tasks`	JSON	nullable	list[SupportTask]
`created_at`	DateTime	default=now	Creation timestamp
`finalized_at`	DateTime	nullable	Finalization timestamp

Indexes: `ix_weekly_programs_plant_year_week`

B.27 `shutdown_events`

Field	Type	Constraints	Description
`shutdown_id`	String(50)	**PK**, default=uuid	Shutdown identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`name`	String(200)	NOT NULL	Shutdown name
`status`	String(20)	default="PLANNED"	ShutdownStatus value
`planned_start`	DateTime	NOT NULL	Planned start datetime
`planned_end`	DateTime	NOT NULL	Planned end datetime
`actual_start`	DateTime	nullable	Actual start
`actual_end`	DateTime	nullable	Actual end
`planned_hours`	Float	default=0.0	Planned duration
`actual_hours`	Float	default=0.0	Actual duration
`work_orders`	JSON	nullable	WO list
`completed_work_orders`	JSON	nullable	Completed WO list
`completion_pct`	Float	default=0.0	Completion percentage
`delay_hours`	Float	default=0.0	Total delay
`delay_reasons`	JSON	nullable	Delay reason list
`created_at`	DateTime	default=now	Creation timestamp

B.28 `moc_requests`

Field	Type	Constraints	Description
`moc_id`	String(50)	**PK**, default=uuid	MoC identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`title`	String(300)	NOT NULL	MoC title
`description`	Text	default=""	MoC description
`category`	String(50)	default="EQUIPMENT_MODIFICATION"	MoCCategory value
`status`	String(20)	default="DRAFT"	MoCStatus value
`risk_level`	String(20)	default="LOW"	RiskLevel value
`requester_id`	String(50)	default=""	Requester ID
`reviewer_id`	String(50)	default=""	Reviewer ID
`approver_id`	String(50)	default=""	Approver ID
`affected_equipment`	JSON	nullable	Affected equipment list
`affected_procedures`	JSON	nullable	Affected procedures
`risk_assessment`	Text	default=""	Risk assessment text
`impact_analysis`	Text	default=""	Impact analysis text
`created_at`	DateTime	default=now	Creation timestamp
`submitted_at`	DateTime	nullable	Submission timestamp
`approved_at`	DateTime	nullable	Approval timestamp
`closed_at`	DateTime	nullable	Closure timestamp

B.29 `spare_part_analyses`

Field	Type	Constraints	Description
`analysis_id`	String(50)	**PK**, default=uuid	Analysis identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`analyzed_at`	DateTime	default=now	Analysis timestamp
`total_parts`	Integer	default=0	Total parts analyzed
`results`	JSON	nullable	list[SparePartAnalysis]
`total_inventory_value`	Float	default=0.0	Total inventory value
`recommended_reduction_pct`	Float	default=0.0	Recommended stock reduction

B.30 `rbi_assessments`

Field	Type	Constraints	Description
`assessment_id`	String(50)	**PK**, default=uuid	Assessment identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`analyzed_at`	DateTime	default=now	Assessment timestamp
`total_equipment`	Integer	default=0	Equipment count assessed
`assessments`	JSON	nullable	list[RBIAssessment]
`high_risk_count`	Integer	default=0	High-risk count
`overdue_count`	Integer	default=0	Overdue inspections

B.31 `reports`

Field	Type	Constraints	Description
`report_id`	String(50)	**PK**, default=uuid	Report identifier
`report_type`	String(30)	NOT NULL	ReportType value
`plant_id`	String(50)	NOT NULL	Plant reference
`period_start`	DateTime	NOT NULL	Report period start
`period_end`	DateTime	NOT NULL	Report period end
`generated_at`	DateTime	default=now	Generation timestamp
`content`	JSON	nullable	Report content
`metadata_json`	JSON	nullable	Report metadata

Indexes: `ix_reports_plant_type`

B.32 `notifications`

Field	Type	Constraints	Description
`notification_id`	String(50)	**PK**, default=uuid	Notification identifier
`notification_type`	String(30)	NOT NULL	NotificationType value
`level`	String(20)	default="INFO"	NotificationLevel value
`plant_id`	String(50)	NOT NULL	Plant reference
`equipment_id`	String(100)	nullable	Equipment reference
`title`	String(300)	NOT NULL	Notification title
`message`	Text	default=""	Notification body
`acknowledged`	Boolean	default=False	Acknowledged flag
`created_at`	DateTime	default=now	Creation timestamp
`acknowledged_at`	DateTime	nullable	Acknowledgement timestamp

B.33 `rca_analyses`

Field	Type	Constraints	Description
`analysis_id`	String(50)	**PK**, default=uuid	Analysis identifier
`event_description`	Text	NOT NULL	Event description
`plant_id`	String(50)	NOT NULL	Plant reference
`equipment_id`	String(100)	nullable	Equipment reference
`level`	String(10)	default="LEVEL_1"	RCA level (1/2/3)
`status`	String(30)	default="OPEN"	RCAStatus value
`team_members`	JSON	nullable	Team member list
`analysis_5w2h`	JSON	nullable	Analysis5W2H data
`cause_effect`	JSON	nullable	CauseEffectDiagram
`evidence_5p`	JSON	nullable	list[Evidence5P]
`solutions`	JSON	nullable	list[Solution]
`created_at`	DateTime	default=now	Creation timestamp
`completed_at`	DateTime	nullable	Completion timestamp

Indexes: `ix_rca_analyses_plant_status`

B.34 `planning_kpi_snapshots`

Field	Type	Constraints	Description
`snapshot_id`	String(50)	**PK**, default=uuid	Snapshot identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`period_start`	Date	NOT NULL	Period start
`period_end`	Date	NOT NULL	Period end
`kpis`	JSON	nullable	list[PlanningKPIValue]
`overall_health`	String(20)	default="UNKNOWN"	HEALTHY, AT_RISK, CRITICAL
`on_target_count`	Integer	default=0	KPIs on target
`below_target_count`	Integer	default=0	KPIs below target
`calculated_at`	DateTime	default=now	Calculation timestamp

Indexes: `ix_planning_kpi_plant_period`

B.35 `de_kpi_snapshots`

Field	Type	Constraints	Description
`snapshot_id`	String(50)	**PK**, default=uuid	Snapshot identifier
`plant_id`	String(50)	NOT NULL	Plant reference
`period_start`	Date	NOT NULL	Period start
`period_end`	Date	NOT NULL	Period end
`kpis`	JSON	nullable	list[DEKPIValue]
`overall_compliance`	Float	default=0.0	Overall compliance score
`program_score`	Float	default=0.0	DE program score
`maturity_level`	String(20)	default="INITIAL"	INITIAL, DEVELOPING, ESTABLISHED, OPTIMIZING
`calculated_at`	DateTime	default=now	Calculation timestamp

Indexes: `ix_de_kpi_plant_period`

B.36 `user_feedback`

Field	Type	Constraints	Description
`feedback_id`	String(50)	**PK**, default=uuid	Feedback identifier
`page`	String(100)	NOT NULL	Page/feature name
`rating`	Integer	NOT NULL	Rating (1-5)
`comment`	Text	default=""	Free-text comment
`created_at`	DateTime	default=now	Submission timestamp

B.37 `fmeca_worksheets`

Field	Type	Constraints	Description
`worksheet_id`	String(50)	**PK**, default=uuid	Worksheet identifier
`equipment_id`	String(100)	indexed	Equipment reference
`equipment_tag`	String(100)	default=""	Equipment TAG
`equipment_name`	String(300)	default=""	Equipment name
`status`	String(20)	default="DRAFT"	FMECAWorksheetStatus value
`current_stage`	String(30)	default="STAGE_1_FUNCTIONS"	FMECAStage value
`rows`	JSON	nullable	Worksheet row data
`stage_completion`	JSON	nullable	Per-stage completion status
`analyst`	String(100)	default=""	Analyst name
`created_at`	DateTime	default=now	Creation timestamp
`completed_at`	DateTime	nullable	Completion timestamp

Section C: API Endpoints

All router endpoints from `api/routers/`.

C.1 Hierarchy (`/hierarchy`)

Method	Path	Request Fields	Response Fields
GET	`/hierarchy/plants`	--	plant_id, name, name_fr, location
POST	`/hierarchy/plants`	plant_id, name, name_fr, name_ar, location	plant_id, name
GET	`/hierarchy/nodes`	?plant_id, ?node_type, ?parent_node_id	node_id, node_type, name, name_fr, code, parent_node_id, level, plant_id, tag, criticality, status, metadata
POST	`/hierarchy/nodes`	node_type, name, code, level, plant_id, parent_node_id, tag, status, metadata	(same as GET node)
GET	`/hierarchy/nodes/{node_id}`	--	node_id, node_type, name, name_fr, code, parent_node_id, level, plant_id, tag, criticality, status, metadata
GET	`/hierarchy/nodes/{node_id}/tree`	--	list of node objects
GET	`/hierarchy/stats`	?plant_id	dict of node_type counts

C.2 Criticality (`/criticality`)

Method	Path	Request Fields	Response Fields
POST	`/criticality/assess`	node_id, criteria_scores, probability, ?method, ?assessed_by	assessment_id, node_id, overall_score, risk_class, criteria_scores, probability, status
GET	`/criticality/{node_id}`	--	assessment_id, node_id, overall_score, risk_class, criteria_scores, probability, status
PUT	`/criticality/{assessment_id}/approve`	--	assessment_id, status
POST	`/criticality/risk-class`	overall_score	risk_class result

C.3 FMEA (`/fmea`)

Method	Path	Request Fields	Response Fields
POST	`/fmea/failure-modes`	functional_failure_id, what, mechanism, cause, failure_consequence, strategy_type, ...	failure_mode_id, what, mechanism, cause, strategy_type
GET	`/fmea/failure-modes/{fm_id}`	--	failure_mode_id, what, mechanism, cause, failure_consequence, strategy_type, failure_effect
GET	`/fmea/failure-modes`	?functional_failure_id	list: failure_mode_id, what, mechanism, cause, strategy_type
POST	`/fmea/validate-combination`	mechanism, cause	validation result
GET	`/fmea/fm-combinations`	?mechanism	valid combinations
POST	`/fmea/rcm-decide`	(RCM decision inputs)	strategy recommendation
POST	`/fmea/functions`	node_id, function_type, description, description_fr	function_id, node_id, function_type
POST	`/fmea/functional-failures`	function_id, failure_type, description, description_fr	failure_id, function_id, failure_type
POST	`/fmea/fmeca/worksheets`	equipment_id, equipment_tag, equipment_name, analyst	worksheet data
GET	`/fmea/fmeca/worksheets/{worksheet_id}`	--	full worksheet data
POST	`/fmea/fmeca/rpn`	severity, occurrence, detection	rpn_value, rpn_category
PUT	`/fmea/fmeca/worksheets/{worksheet_id}/run-decisions`	--	updated worksheet with RCM decisions
GET	`/fmea/fmeca/worksheets/{worksheet_id}/summary`	--	summary statistics

C.4 Tasks (`/tasks`)

Method	Path	Request Fields	Response Fields
POST	`/tasks/`	name, task_type, constraint, frequency_value, frequency_unit, ...	task_id, name, task_type, status
GET	`/tasks/{task_id}`	--	task_id, name, task_type, constraint, frequency_value, frequency_unit, status, labour_resources, material_resources
GET	`/tasks/`	?failure_mode_id, ?status	list: task_id, name, task_type, status
POST	`/tasks/link-fm/{task_id}/{fm_id}`	--	task_id, failure_mode_id
POST	`/tasks/validate-name`	name, ?task_type	validation result
POST	`/tasks/validate-wp-name`	name	validation result

C.5 Work Packages (`/work-packages`)

Method	Path	Request Fields	Response Fields
POST	`/work-packages/`	name, code, node_id, frequency_value, frequency_unit, constraint, work_package_type	work_package_id, name, code, status
GET	`/work-packages/{wp_id}`	--	work_package_id, name, code, node_id, frequency_value, frequency_unit, constraint, work_package_type, status, allocated_tasks, labour_summary
GET	`/work-packages/`	?node_id, ?status	list: work_package_id, name, code, status
PUT	`/work-packages/{wp_id}/approve`	--	work_package_id, status
POST	`/work-packages/group`	items (list of tasks)	grouped work packages
POST	`/work-packages/{wp_id}/work-instruction`	equipment_name, equipment_tag, tasks	work instruction document

C.6 SAP (`/sap`)

Method	Path	Request Fields	Response Fields
POST	`/sap/generate-upload`	plant_code, maintenance_plan, maintenance_items, task_lists	package_id, status
GET	`/sap/uploads/{package_id}`	--	package_id, plant_code, status, generated_at, maintenance_plan, maintenance_items, task_lists
GET	`/sap/uploads`	?plant_code	list: package_id, plant_code, status
PUT	`/sap/uploads/{package_id}/approve`	--	approval result
POST	`/sap/validate-transition`	entity_type, current_state, target_state	validation result
GET	`/sap/mock/{transaction}`	--	mock SAP data

C.7 Analytics (`/analytics`)

Method	Path	Request Fields	Response Fields
POST	`/analytics/health-score`	node_id, plant_id, equipment_tag, risk_class, ?pending_backlog_hours, ?capacity_hours_per_week, ?total_failure_modes, ?fm_with_strategy, ?active_alerts, ?critical_alerts, ?planned_wo, ?executed_on_time	AssetHealthScore
POST	`/analytics/kpis`	plant_id, ?failure_dates, ?total_period_hours, ?total_downtime_hours	KPIMetrics
POST	`/analytics/weibull-fit`	failure_intervals	WeibullParameters
POST	`/analytics/weibull-predict`	equipment_id, equipment_tag, failure_intervals, current_age_days, ?confidence_level	FailurePrediction
POST	`/analytics/variance-detect`	snapshots	list of PlantVarianceAlert
GET	`/analytics/variance-alerts`	--	list: alert_id, plant_id, metric_name, z_score, variance_level

C.8 Capture (`/capture`)

Method	Path	Request Fields	Response Fields
POST	`/capture/`	technician_id, capture_type, language, raw_text, raw_voice_text, images, equipment_tag_manual, location_hint	capture result with work request
GET	`/capture/`	--	list: capture_id, technician_id, capture_type, language, equipment_tag_manual, created_at
GET	`/capture/{capture_id}`	--	capture_id, technician_id, capture_type, language, raw_text, raw_voice_text, equipment_tag_manual, location_hint, created_at

C.9 Work Requests (`/work-requests`)

Method	Path	Request Fields	Response Fields
GET	`/work-requests/`	?status	list: request_id, status, equipment_tag, equipment_confidence, ai_classification, created_at
GET	`/work-requests/{request_id}`	--	request_id, source_capture_id, status, equipment_id, equipment_tag, equipment_confidence, problem_description, ai_classification, spare_parts, image_analysis, validation, created_at
PUT	`/work-requests/{request_id}/validate`	action (APPROVE/REJECT/MODIFY), ?modifications	validation result
POST	`/work-requests/{request_id}/classify`	--	classification result

C.10 Planner (`/planner`)

Method	Path	Request Fields	Response Fields
POST	`/planner/{work_request_id}/recommend`	--	recommendation result
GET	`/planner/recommendations/{recommendation_id}`	--	recommendation_id, work_request_id, planner_action, ai_confidence, resource_analysis, scheduling_suggestion, risk_assessment, generated_at
PUT	`/planner/recommendations/{recommendation_id}/action`	action (APPROVE/MODIFY/ESCALATE/DEFER), ?modifications	action result

C.11 Backlog (`/backlog`)

Method	Path	Request Fields	Response Fields
GET	`/backlog/`	?status, ?priority, ?equipment_tag	backlog items list
POST	`/backlog/add/{work_request_id}`	--	added backlog item
POST	`/backlog/optimize`	?plant_id, ?period_days	optimization result
GET	`/backlog/optimizations/{optimization_id}`	--	optimization_id, plant_id, status, total_items, stratification, work_packages, schedule, alerts, generated_at
PUT	`/backlog/optimizations/{optimization_id}/approve`	--	approval result
GET	`/backlog/schedule`	--	current schedule

C.12 Scheduling (`/scheduling`)

Method	Path	Request Fields	Response Fields
POST	`/scheduling/programs`	?plant_id, ?week_number, ?year	program data
GET	`/scheduling/programs`	?plant_id, ?status	list of programs
GET	`/scheduling/programs/{program_id}`	--	program_id, plant_id, week_number, year, status, total_hours, work_packages, resource_slots, conflicts, support_tasks, created_at, finalized_at
PUT	`/scheduling/programs/{program_id}/finalize`	--	finalization result
PUT	`/scheduling/programs/{program_id}/activate`	--	activation result
PUT	`/scheduling/programs/{program_id}/complete`	--	completion result
GET	`/scheduling/programs/{program_id}/gantt`	--	Gantt chart data
GET	`/scheduling/programs/{program_id}/gantt/export`	--	Excel file download

C.13 Reliability (`/reliability`)

Method	Path	Request Fields	Response Fields
POST	`/reliability/spare-parts/analyze`	?plant_id, parts	SparePartOptimizationResult
POST	`/reliability/shutdowns`	plant_id, name, planned_start, planned_end, work_orders	shutdown data
GET	`/reliability/shutdowns/{shutdown_id}`	--	full shutdown data
PUT	`/reliability/shutdowns/{shutdown_id}/start`	--	updated shutdown
PUT	`/reliability/shutdowns/{shutdown_id}/complete`	--	completed shutdown
POST	`/reliability/moc`	plant_id, title, description, category, requester_id, affected_equipment, risk_level	MoC data
GET	`/reliability/moc`	?plant_id, ?status	list of MoCs
GET	`/reliability/moc/{moc_id}`	--	full MoC data
PUT	`/reliability/moc/{moc_id}/advance`	action, ...	updated MoC
POST	`/reliability/ocr/analyze`	equipment_id, failure_rate, mttr_hours, cost_per_failure, cost_per_pm, current_pm_interval_days	OCRAnalysisResult
POST	`/reliability/jackknife/analyze`	?plant_id, equipment_data	JackKnifeResult
POST	`/reliability/pareto/analyze`	?plant_id, ?metric_type, records	ParetoResult
POST	`/reliability/lcc/calculate`	equipment_id, acquisition_cost, installation_cost, annual_operating_cost,	LCCResult

Method	Path	Request Fields	Response Fields
		annual_maintenance_cost, expected_life_years, discount_rate, salvage_value	
POST	`/reliability/rbi/assess`	?plant_id, equipment_list	RBIResult

C.14 Reporting (`/reporting`)

Method	Path	Request Fields	Response Fields
POST	`/reporting/reports/weekly`	?plant_id, week/week_number, year, ...	WeeklyReport
POST	`/reporting/reports/monthly`	?plant_id, month, year, ...	MonthlyKPIReport
POST	`/reporting/reports/quarterly`	?plant_id, quarter, year, ...	QuarterlyReviewReport
GET	`/reporting/reports`	?plant_id, ?report_type	list of reports
GET	`/reporting/reports/{report_id}`	--	full report data
POST	`/reporting/de-kpis/calculate`	(DEKPIInput fields)	DEKPIs
POST	`/reporting/de-kpis/program-health`	(DE KPI data)	DEProgramHealth
POST	`/reporting/notifications/generate`	plant_id, config	NotificationResult
GET	`/reporting/notifications`	?plant_id, ?level	list of notifications
PUT	`/reporting/notifications/{notification_id}/ack`	--	acknowledgement result
POST	`/reporting/import/validate`	source, rows	ImportResult
POST	`/reporting/export`	export_type, ...	ExportResult
POST	`/reporting/cross-module/analyze`	plant_id, ...	CrossModuleSummary

C.15 RCA (`/rca`)

Method	Path	Request Fields	Response Fields
POST	`/rca/analyses`	event_description, ?plant_id, ?equipment_id, ?max_consequence, ?frequency, ?team_members	RCA analysis data
GET	`/rca/analyses`	?plant_id, ?status	list of RCA analyses
GET	`/rca/analyses/summary`	?plant_id	summary statistics
GET	`/rca/analyses/{analysis_id}`	--	full RCA analysis
POST	`/rca/analyses/{analysis_id}/5w2h`	what, when, where, who, why, how, how_much	5W+2H result
PUT	`/rca/analyses/{analysis_id}/advance`	status	status transition result
POST	`/rca/planning-kpis/calculate`	(PlanningKPIInput fields)	PlanningKPIs
GET	`/rca/planning-kpis`	?plant_id	list of snapshots
POST	`/rca/de-kpis/calculate`	(DEKPIInput fields)	DEKPIs
GET	`/rca/de-kpis`	?plant_id	list of DE KPI snapshots

C.16 Dashboard (`/dashboard`)

Method	Path	Request Fields	Response Fields
GET	`/dashboard/executive/{plant_id}`	--	plant_id, total_reports, recent_reports, total_notifications, critical_alerts, recent_notifications
GET	`/dashboard/kpi-summary/{plant_id}`	--	plant_id, has_data, report
GET	`/dashboard/alerts/{plant_id}`	--	plant_id, total_active, alerts

C.17 Admin (`/admin`)

Method	Path	Request Fields	Response Fields
POST	`/admin/seed-database`	--	seed result
GET	`/admin/audit-log`	?entity_type, ?limit	list: id, entity_type, entity_id, action, user, timestamp
GET	`/admin/stats`	--	plants, hierarchy_nodes, total_nodes
DELETE	`/admin/reset-database`	--	status confirmation
GET	`/admin/agent-status`	--	agent status data
POST	`/admin/feedback`	page, rating, comment	feedback_id, status
GET	`/admin/feedback`	?page, ?limit	list: feedback_id, page, rating, comment, created_at

---

Section D: KPI Definitions

D.1 Planning KPIs (11 KPIs -- GFSN REF-14 S8)

Source: `tools/engines/planning_kpi_engine.py`

#	KPI Name	Formula	GFSN Target	Unit
1	**WO Completion Rate**	`wo_completed / wo_planned * 100`	>= 90%	%
2	**Man-hour Compliance**	`manhours_actual / manhours_planned * 100`	85--115%	%
3	**PM Plan Compliance**	`pm_executed / pm_planned * 100`	>= 95%	%
4	**Backlog Weeks**	`backlog_hours / weekly_capacity_hours`	<= 4 weeks	weeks
5	**Reactive Work %**	`corrective_count / total_wo * 100`	<= 20%	%
6	**Schedule Adherence**	`schedule_compliance_executed / schedule_compliance_planned * 100`	>= 85%	%
7	**Release Horizon**	`release_horizon_days` (direct input)	<= 7 days	days
8	**Pending Notices %**	`pending_notices / total_notices * 100`	<= 15%	%
9	**Scheduled Capacity**	`scheduled_capacity_hours / total_capacity_hours * 100`	80--95%	%
10	**Proactive Work %**	`proactive_wo / total_wo * 100`	>= 70%	%
11	**Planning Efficiency**	`planned_wo / total_wo * 100`	>= 85%	%

Overall Health Classification:

> - > *= 9 KPIs on target:* `HEALTHY`
> - *6-8 KPIs on target:* `AT_RISK`
> - *< 6 KPIs on target:* `CRITICAL`

D.2 Defect Elimination KPIs (5 KPIs -- GFSN REF-15)

Source: `tools/engines/rca_engine.py` (compute_de_kpis), `tools/engines/de_kpi_engine.py`

#	KPI Name	Formula	Target	Unit
1	**Event Reporting Compliance**	`events_reported / events_required * 100`	>= 95%	%
2	**Meeting Compliance**	`meetings_held / meetings_required * 100`	>= 90%	%
3	**Implementation Progress**	`actions_implemented / actions_planned * 100`	>= 80%	%
4	**Savings Effectiveness**	`savings_achieved / savings_target * 100`	>= 70%	%
5	**Frequency Reduction**	`(failures_previous - failures_current) / failures_previous * 100`	>= 10%	%

DE Program Maturity Levels (from `DEKPIEngine.assess_program_health`):

- Score >= 80: `OPTIMIZING`
- Score >= 60: `ESTABLISHED`
- Score >= 40: `DEVELOPING`

- Score < 40: `INITIAL`

D.3 Core Reliability KPIs (7 KPIs -- ISO 55002 S9.1)

Source: `tools/engines/kpi_engine.py`

#	KPI Name	Formula	Description
1	**MTBF**	`sum(intervals between failure dates) / count(intervals)`	Mean Time Between Failures in days. Requires >= 2 failure events.
2	**MTTR**	`sum(repair_durations) / count(repairs)`	Mean Time To Repair in hours. Only positive durations counted.
3	**Availability**	`(total_period_hours - total_downtime_hours) / total_period_hours * 100`	Uptime percentage (0-100%).
4	**OEE**	`(availability/100) * (performance/100) * (quality/100) * 100`	Overall Equipment Effectiveness. Performance and quality default to 100% in MVP.
5	**Schedule Compliance**	`executed_on_time / planned_count * 100`	Percentage of WOs completed on schedule.
6	**PM Compliance**	`pm_executed / pm_planned * 100`	Preventive maintenance execution rate.
7	**Reactive Ratio**	`corrective_count / total_count * 100`	Percentage of work that is corrective/reactive.

D.4 Asset Health Index (5 Dimensions -- REF-12 Rec 4)

Source: `tools/engines/health_score_engine.py`

#	Dimension	Default Weight	Score Calculation	Details
1	**Criticality**	0.25	Risk class mapping: I_LOW=90, II_MEDIUM=70, III_HIGH=40, IV_CRITICAL=15	Inverse: higher criticality = lower health score
2	**Backlog Pressure**	0.20	`100 * (1 - weeks_of_backlog / 8)` where weeks = pending_hours / capacity_per_week	0 pending = 100 (healthy); >= 8 weeks backlog = 0
3	**Strategy Coverage**	0.25	`100 * fm_with_strategy / total_failure_modes`	Percentage of failure modes covered by approved strategy
4	**Condition Status**	0.15	`100 * (1 - (active_alerts + critical_alerts * 2) / 10)`	Critical alerts weighted 3x; max threshold = 10
5	**Execution Compliance**	0.15	`100 * executed_on_time / planned_wo`	On-time work order execution rate

Composite Score: `sum(dimension_score * weight) / sum(weights)`

Health Classification:

> - > *= 75: `HEALTHY`*
> - > *= 50: `AT_RISK`*
> - > *0: `CRITICAL`*
> - *0: `UNKNOWN`*

Trend Detection: Delta > 5 = `IMPROVING`; Delta < -5 = `DEGRADING`; else `STABLE`

Section E: 72 Valid Failure Mode Combinations

The authoritative lookup table from `VALID_FM_COMBINATIONS` in `tools/models/schemas.py`. Source: SRC-09 -- "Failure Modes (Mechanism + Cause).xlsx". Any `FailureMode` must use one of these 72 mechanism+cause pairs.

#	Mechanism	Cause
1	ARCS	BREAKDOWN_IN_INSULATION
2	BLOCKS	CONTAMINATION
3	BLOCKS	EXCESSIVE_PARTICLE_SIZE
4	BLOCKS	INSUFFICIENT_FLUID_VELOCITY
5	BREAKS_FRACTURE_SEPARATES	CYCLIC_LOADING
6	BREAKS_FRACTURE_SEPARATES	MECHANICAL_OVERLOAD
7	BREAKS_FRACTURE_SEPARATES	THERMAL_OVERLOAD
8	CORRODES	BIO_ORGANISMS
9	CORRODES	CHEMICAL_ATTACK
10	CORRODES	CORROSIVE_ENVIRONMENT
11	CORRODES	CREVICE
12	CORRODES	DISSIMILAR_METALS_CONTACT
13	CORRODES	EXPOSURE_TO_ATMOSPHERE
14	CORRODES	EXPOSURE_TO_HIGH_TEMP_CORROSIVE
15	CORRODES	EXPOSURE_TO_HIGH_TEMP
16	CORRODES	EXPOSURE_TO_LIQUID_METAL
17	CORRODES	POOR_ELECTRICAL_CONNECTIONS
18	CORRODES	POOR_ELECTRICAL_INSULATION
19	CRACKS	AGE
20	CRACKS	CYCLIC_LOADING
21	CRACKS	EXCESSIVE_TEMPERATURE
22	CRACKS	HIGH_TEMP_CORROSIVE
23	CRACKS	IMPACT_SHOCK_LOADING
24	CRACKS	THERMAL_STRESSES
25	DEGRADES	AGE
26	DEGRADES	CHEMICAL_ATTACK

#	Mechanism	Cause
27	DEGRADES	CHEMICAL_REACTION
28	DEGRADES	CONTAMINATION
29	DEGRADES	ELECTRICAL_ARCING
30	DEGRADES	ENTRAINED_AIR
31	DEGRADES	EXCESSIVE_TEMPERATURE
32	DEGRADES	RADIATION
33	DISTORTS	IMPACT_SHOCK_LOADING
34	DISTORTS	MECHANICAL_OVERLOAD
35	DISTORTS	OFF_CENTER_LOADING
36	DISTORTS	USE
37	DRIFTS	EXCESSIVE_TEMPERATURE
38	DRIFTS	IMPACT_SHOCK_LOADING
39	DRIFTS	STRAY_CURRENT
40	DRIFTS	UNEVEN_LOADING
41	DRIFTS	USE
42	EXPIRES	AGE
43	IMMOBILISED	CONTAMINATION
44	IMMOBILISED	LACK_OF_LUBRICATION
45	LOOSES_PRELOAD	CREEP
46	LOOSES_PRELOAD	EXCESSIVE_TEMPERATURE
47	LOOSES_PRELOAD	VIBRATION
48	OPEN_CIRCUIT	ELECTRICAL_OVERLOAD
49	OVERHEATS_MELTS	CONTAMINATION
50	OVERHEATS_MELTS	ELECTRICAL_OVERLOAD
51	OVERHEATS_MELTS	LACK_OF_LUBRICATION
52	OVERHEATS_MELTS	MECHANICAL_OVERLOAD
53	OVERHEATS_MELTS	RELATIVE_MOVEMENT

#	Mechanism	Cause
54	OVERHEATS_MELTS	RUBBING
55	SEVERS	ABRASION
56	SEVERS	IMPACT_SHOCK_LOADING
57	SEVERS	MECHANICAL_OVERLOAD
58	SHORT_CIRCUITS	BREAKDOWN_IN_INSULATION
59	SHORT_CIRCUITS	CONTAMINATION
60	THERMALLY_OVERLOADS	MECHANICAL_OVERLOAD
61	THERMALLY_OVERLOADS	OVERCURRENT
62	WASHES_OFF	EXCESSIVE_FLUID_VELOCITY
63	WASHES_OFF	USE
64	WEARS	BREAKDOWN_OF_LUBRICATION
65	WEARS	ENTRAINED_AIR
66	WEARS	EXCESSIVE_FLUID_VELOCITY
67	WEARS	IMPACT_SHOCK_LOADING
68	WEARS	LOW_PRESSURE
69	WEARS	LUBRICANT_CONTAMINATION
70	WEARS	MECHANICAL_OVERLOAD
71	WEARS	METAL_TO_METAL_CONTACT
72	WEARS	RELATIVE_MOVEMENT

Section F: Excel Template Schemas (14 Templates)

All `.xlsx` data-loading templates generated by `templates/generate_templates.py`. Each template includes OCP green headers, frozen panes, data validation dropdowns, and 3-5 realistic phosphate example rows.

F.1 Template 01: `01_equipment_hierarchy.xlsx`

Plant and equipment hierarchy nodes (18 fields).

Field	Type	Validation	Description
`plant_id`	String	required	Plant identifier (e.g., OCP-JFC1)
`plant_name`	String	required	Plant name
`area_code`	String	required	Area code (e.g., BRY)
`area_name`	String	required	Area name
`system_code`	String	required	System code
`system_name`	String	required	System name
`equipment_tag`	String	required	Equipment tag (e.g., BRY-SAG-ML-001)
`equipment_description`	String	required	Equipment description
`equipment_type`	String	dropdown: EquipmentCategory	Equipment type
`manufacturer`	String	-	OEM manufacturer
`model`	String	-	Model number
`serial_number`	String	-	Serial number
`power_kw`	Float	-	Rated power (kW)
`weight_kg`	Float	-	Weight (kg)
`criticality`	String	dropdown: EquipmentCriticality	Criticality grade
`status`	String	dropdown: EquipmentStatus	Equipment status
`sap_func_loc`	String	-	SAP functional location
`installation_date`	Date	YYYY-MM-DD	Installation date

F.2 Template 02: `02_criticality_assessment.xlsx`

Criticality assessment scores (14 fields).

Field	Type	Validation	Description
`equipment_tag`	String	required	Equipment tag reference
`method`	String	dropdown: CriticalityMethod	R8 or GFSN method
`safety`	Integer	1-5	Safety consequence score
`health`	Integer	1-5	Health consequence score
`environment`	Integer	1-5	Environment consequence score
`production`	Integer	1-5	Production consequence score
`operating_cost`	Integer	1-5	Operating cost score
`capital_cost`	Integer	1-5	Capital cost score
`schedule`	Integer	1-5	Schedule impact score
`revenue`	Integer	1-5	Revenue impact score
`communications`	Integer	1-5	Communications impact score
`compliance`	Integer	1-5	Compliance impact score
`reputation`	Integer	1-5	Reputation impact score
`probability`	Integer	1-5	Probability of occurrence

F.3 Template 03: `03_failure_modes.xlsx`

FMEA failure modes and causes (15 fields + reference sheet).

Field	Type	Validation	Description
`equipment_tag`	String	required	Equipment tag reference
`function_description`	String	required	Function description
`function_type`	String	dropdown: FunctionType	PRIMARY, SECONDARY, PROTECTIVE
`failure_type`	String	dropdown: FailureType	TOTAL or PARTIAL
`what_component`	String	required	Component that fails (uppercase, singular)
`mechanism`	String	dropdown: Mechanism	Failure mechanism (18 values)
`cause`	String	dropdown: Cause	Failure cause (must be valid 72-combo pair)
`failure_pattern`	String	dropdown: FailurePattern	Nowlan-Heap pattern (A-F)
`failure_consequence`	String	dropdown: FailureConsequence	Consequence classification
`evidence`	String	required	Observable evidence of failure
`downtime_hours`	Float	>=0	Estimated downtime
`detection_method`	String	-	How failure is detected
`rpn_severity`	Integer	1-10	RPN severity score
`rpn_occurrence`	Integer	1-10	RPN occurrence score
`rpn_detection`	Integer	1-10	RPN detection score

Additional sheet: "Valid FM Combinations" -- all 72 valid Mechanism+Cause pairs.

F.4 Template 04: `04_maintenance_tasks.xlsx`

Maintenance task definitions (multi-sheet: Tasks, Task_Labour, Task_Materials, Task_Tools).

Sheet 1: Tasks (14 fields)

Field	Type	Validation	Description
`task_id`	String	required, PK	Unique task identifier
`task_name`	String	required, max 72 chars	Task name (SAP compatible)
`task_name_fr`	String	-	Task name (French)
`task_type`	String	dropdown: TaskType	INSPECT, CHECK, TEST, LUBRICATE, CLEAN, REPLACE, REPAIR, CALIBRATE
`constraint`	String	dropdown: TaskConstraint	ONLINE, OFFLINE, TEST_MODE
`access_time_hours`	Float	>=0	Access/setup time
`budgeted_as`	String	dropdown: BudgetType	NOT_BUDGETED, REPAIR, REPLACE
`budgeted_life`	Float	-	Budgeted life value
`budgeted_life_time_units`	String	-	Time-based units
`budgeted_life_operational_units`	String	-	Operational units
`consequences`	String	-	Failure consequences
`justification`	String	-	Task justification
`origin`	String	-	Task origin reference
`notes`	String	-	Additional notes

> ***Note:*** *Tasks do NOT have frequency or acceptable limits. Those belong to the maintenance strategy (Template 14).*

Sheet 2: Task_Labour (8 fields)

Field	Type	Validation	Description
`labour_id`	String	required, PK	Unique labour line ID
`task_id`	String	required, FK → Tasks	Parent task reference
`worker_id`	String	-	Worker identifier
`specialty`	String	dropdown: LabourSpecialty	Worker specialty
`quantity`	Integer	>=1	Number of workers
`hours_per_person`	Float	>=0	Hours per person
`hourly_rate_usd`	Float	>=0	Hourly rate (USD)
`company`	String	-	Contractor company

Sheet 3: Task_Materials (10 fields)

Field	Type	Validation	Description
`material_line_id`	String	required, PK	Unique material line ID
`task_id`	String	required, FK → Tasks	Parent task reference
`material_code`	String	required	SAP material code
`description`	String	required	Material description
`manufacturer`	String	-	Material manufacturer
`part_number`	String	-	Part number
`quantity`	Float	>=0	Quantity required
`unit_of_measure`	String	dropdown: UnitOfMeasure	Unit
`unit_price_usd`	Float	>=0	Unit price (USD)
`equipment_bom_ref`	String	-	BOM reference

Sheet 4: Task_Tools (5 fields)

Field	Type	Validation	Description
`tool_line_id`	String	required, PK	Unique tool line ID
`task_id`	String	required, FK → Tasks	Parent task reference
`item_type`	String	-	TOOL or SPECIAL_EQUIPMENT
`tool_code`	String	required	Tool code
`description`	String	-	Tool description

F.5 Template 05: `05_work_packages.xlsx`

Work package groupings (11 fields).

Field	Type	Validation	Description
`wp_name`	String	required, max 40, ALL CAPS	Work package name
`wp_code`	String	required	WP code
`equipment_tag`	String	required	Equipment tag reference
`frequency_value`	Float	>=0	Interval value
`frequency_unit`	String	dropdown: FrequencyUnit	Interval unit
`constraint`	String	dropdown: WPConstraint	ONLINE or OFFLINE
`wp_type`	String	dropdown: WPType	STANDALONE, SUPPRESSIVE, SEQUENTIAL
`access_time_hours`	Float	>=0	Access/setup time
`task_ids_csv`	String	required	Comma-separated task IDs
`estimated_total_hours`	Float	>=0	Total estimated hours
`crew_size`	Integer	>=1	Crew size

F.6 Template 06: `06_work_order_history.xlsx`

Historical work order data (16 fields).

Field	Type	Validation	Description
`wo_id`	String	required	Work order ID
`order_type`	String	dropdown: WorkOrderType	Order type
`equipment_tag`	String	required	Equipment tag
`sap_func_loc`	String	-	SAP functional location
`priority`	String	dropdown: Priority	Priority level
`description`	String	required	WO description
`created_date`	Date	YYYY-MM-DD	Creation date
`planned_start`	Date	YYYY-MM-DD	Planned start
`planned_end`	Date	YYYY-MM-DD	Planned end
`actual_start`	Date	YYYY-MM-DD	Actual start
`actual_end`	Date	YYYY-MM-DD	Actual end
`duration_hours`	Float	>=0	Duration (hours)
`status`	String	dropdown: WOHistoryStatus	WO status
`cost_labour_usd`	Float	>=0	Labour cost
`cost_materials_usd`	Float	>=0	Materials cost
`failure_found`	String	-	Failure found (Y/N or description)

F.7 Template 07: `07_spare_parts_inventory.xlsx`

Spare parts catalog (16 fields).

Field	Type	Validation	Description
`material_code`	String	required	SAP material code
`description`	String	required	Part description
`manufacturer`	String	-	Manufacturer
`part_number`	String	-	Part number
`ved_class`	String	dropdown: V,E,D	VED criticality
`fsn_class`	String	dropdown: F,S,N	FSN consumption
`abc_class`	String	dropdown: A,B,C	ABC cost class
`quantity_on_hand`	Integer	>=0	Current stock
`min_stock`	Integer	>=0	Minimum stock level
`max_stock`	Integer	>=0	Maximum stock level
`reorder_point`	Integer	>=0	Reorder point
`lead_time_days`	Integer	>=0	Lead time (days)
`unit_cost_usd`	Float	>=0	Unit cost (USD)
`unit_of_measure`	String	dropdown: UnitOfMeasure	Unit
`applicable_equipment_csv`	String	-	Comma-separated equipment tags
`warehouse_location`	String	-	Storage location

F.8 Template 08: `08_shutdown_calendar.xlsx`

Planned shutdown schedule (9 fields).

Field	Type	Validation	Description
`plant_id`	String	required	Plant identifier
`shutdown_name`	String	required	Shutdown name
`shutdown_type`	String	dropdown: ShutdownType	MAJOR, MINOR, TURNAROUND
`planned_start`	Date	YYYY-MM-DD	Planned start
`planned_end`	Date	YYYY-MM-DD	Planned end
`planned_hours`	Float	>=0	Planned duration (hours)
`areas_csv`	String	-	Affected areas (comma-separated)
`description`	String	-	Shutdown description
`work_orders_csv`	String	-	Linked WO IDs (comma-separated)

F.9 Template 09: `09_workforce.xlsx`

Workforce and specialty data (9 fields).

Field	Type	Validation	Description
`worker_id`	String	required	Worker identifier
`name`	String	required	Worker name
`specialty`	String	dropdown: LabourSpecialty	Worker specialty
`shift`	String	dropdown: ShiftType	Shift assignment
`plant_id`	String	required	Plant identifier
`available`	Boolean	TRUE/FALSE	Availability flag
`certifications_csv`	String	-	Certifications (comma-separated)
`phone`	String	-	Phone number
`email`	String	-	Email address

F.10 Template 10: `10_field_capture.xlsx`

Field observation records (7 fields).

Field	Type	Validation	Description
`technician_id`	String	required	Technician ID
`capture_type`	String	dropdown: CaptureType	Capture type
`language`	String	dropdown: Language	Input language
`equipment_tag`	String	required	Equipment tag
`location_hint`	String	-	Location hint
`raw_text`	String	required	Raw observation text
`timestamp`	DateTime	ISO 8601	Capture timestamp

F.11 Template 11: `11_rca_events.xlsx`

Root cause analysis events (10 fields).

Field	Type	Validation	Description
`event_description`	String	required	Event description
`plant_id`	String	required	Plant identifier
`equipment_tag`	String	required	Equipment tag
`level`	String	dropdown: RCALevel	GREEN, YELLOW, RED
`max_consequence`	String	-	Maximum consequence type
`frequency`	String	-	Event frequency
`event_date`	Date	YYYY-MM-DD	Event date
`downtime_hours`	Float	>=0	Downtime caused
`production_loss_tonnes`	Float	>=0	Production loss (tonnes)
`direct_cost_usd`	Float	>=0	Direct cost (USD)

F.12 Template 12: `12_planning_kpi_input.xlsx`

Planning KPI input data (22 fields).

Field	Type	Validation	Description
`plant_id`	String	required	Plant identifier
`period_start`	Date	YYYY-MM-DD	Period start date
`period_end`	Date	YYYY-MM-DD	Period end date
`wo_planned`	Integer	>=0	Work orders planned
`wo_completed`	Integer	>=0	Work orders completed
`manhours_planned`	Float	>=0	Man-hours planned
`manhours_actual`	Float	>=0	Man-hours actual
`pm_planned`	Integer	>=0	PM orders planned
`pm_executed`	Integer	>=0	PM orders executed
`backlog_hours`	Float	>=0	Backlog hours
`weekly_capacity_hours`	Float	>=0	Weekly capacity
`corrective_count`	Integer	>=0	Corrective WO count
`total_wo`	Integer	>=0	Total WO count
`schedule_compliance_planned`	Integer	>=0	Planned schedule items
`schedule_compliance_executed`	Integer	>=0	Executed schedule items
`release_horizon_days`	Integer	>=0	Release horizon (days)
`pending_notices`	Integer	>=0	Pending notices
`total_notices`	Integer	>=0	Total notices
`scheduled_capacity_hours`	Float	>=0	Scheduled capacity
`total_capacity_hours`	Float	>=0	Total capacity
`proactive_wo`	Integer	>=0	Proactive WO count
`planned_wo`	Integer	>=0	Planned WO count

F.13 Template 13: `13_de_kpi_input.xlsx`

Defect elimination KPI data (13 fields).

Field	Type	Validation	Description
`plant_id`	String	required	Plant identifier
`period_start`	Date	YYYY-MM-DD	Period start date
`period_end`	Date	YYYY-MM-DD	Period end date
`events_reported`	Integer	>=0	Events reported
`events_required`	Integer	>=0	Events requiring analysis
`meetings_held`	Integer	>=0	DE meetings held
`meetings_required`	Integer	>=0	DE meetings required
`actions_implemented`	Integer	>=0	Actions implemented
`actions_planned`	Integer	>=0	Actions planned
`savings_achieved`	Float	>=0	Savings achieved (USD)
`savings_target`	Float	>=0	Savings target (USD)
`failures_current`	Integer	>=0	Current period failures
`failures_previous`	Integer	>=0	Previous period failures

F.14 Template 14: `14_maintenance_strategy.xlsx`

Maintenance strategy linking failure modes to tasks with intervals and limits (31 fields + 2 reference sheets).

> *This is the **core strategy template** that connects the FMEA analysis (Template 03) to maintenance tasks (Template 04). Each row represents a strategy record: one failure mode → one or two maintenance tasks, with frequency, acceptable limits, and justification.*

Field	Type	Validation	Description
`strategy_id`	String	required	Unique strategy identifier
`equipment_tag`	String	required	Equipment tag reference
`maintainable_item`	String	required	Maintainable item name
`function_and_failure`	String	required	Function and failure description
`what`	String	required	FM "what" component
`mechanism`	String	dropdown: Mechanism	Failure mechanism (18 values)
`cause`	String	dropdown: Cause	Failure cause (72-combo validated)
`status`	String	dropdown: ACTIVE, DRAFT, ELIMINATED	Strategy status
`tactics_type`	String	dropdown: StrategyType	CBM, TBM, FBM, RTF, FFI, REDESIGN
`primary_task_id`	String	required, FK → Tasks	Primary task reference
`primary_task_interval`	Float	>=0	Primary task interval value
`operational_units`	String	-	Operational interval units
`time_units`	String	dropdown: FrequencyUnit	Time-based interval units
`primary_task_acceptable_limits`	String	-	Measurable acceptance criteria
`primary_task_conditional_comments`	String	-	Conditional notes
`primary_task_constraint`	String	dropdown: TaskConstraint	ONLINE, OFFLINE, TEST_MODE
`primary_task_task_type`	String	dropdown: TaskType	Primary task type
`primary_task_access_time`	Float	>=0	Primary task access time
`secondary_task_id`	String	FK → Tasks	Secondary task reference

Field	Type	Validation	Description
`secondary_task_constraint`	String	dropdown: TaskConstraint	Secondary task constraint
`secondary_task_task_type`	String	dropdown: TaskType	Secondary task type
`secondary_task_access_time`	Float	>=0	Secondary task access time
`secondary_task_comments`	String	-	Secondary task notes
`budgeted_as`	String	dropdown: BudgetType	NOT_BUDGETED, REPAIR, REPLACE
`budgeted_life`	Float	-	Budgeted life value
`budgeted_life_time_units`	String	-	Budgeted life time units
`budgeted_life_operational_units`	String	-	Budgeted life operational units
`existing_task`	String	-	Reference to existing task (if any)
`justification_category`	String	dropdown: JustificationCategory	Justification category
`justification`	String	-	Strategy justification text
`notes`	String	-	Additional notes

Additional sheet 1: "Valid FM Combinations" -- all 72 valid Mechanism+Cause pairs.

Additional sheet 2: "Strategy Type Rules" -- 6 maintenance strategy types with selection criteria:

Strategy	When to Use
`CBM`	Condition-Based: detectable P-F interval, feasible monitoring
`TBM`	Time-Based: age-related pattern, known useful life
`FBM`	Failure-Based: redesign not justified, low consequence RTF
`RTF`	Run-to-Failure: non-critical, no safety/environmental consequence
`FFI`	Failure-Finding Interval: hidden failure, protective function
`REDESIGN`	Redesign: no applicable task, unacceptable risk

End of variable listing.